

The Finite-Compression Determinant Infimum Characterizes Amenability

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Abstract

Austin asks whether a countable group must be amenable if his finite-subset determinant lower bound is always sharp for positive definite functions of the form $\phi = \tau(a^*(\cdot)a)$ with $a \in \mathbf{C}[\Gamma]$. We give an affirmative answer. The main ingredient is a quantitative form of the operator Jensen inequality for compressions of log. Applied to $a = 1 + t \sum_{s \in S} s$, it bounds the logarithmic determinant gap below by a fixed multiple of the normalized S -boundary of the compressing set. Thus sharpness for these elementary group-ring elements yields the Følner criterion. The same estimate gives a direct quantitative answer to the following problem in the source for this family of elements.

1 Source question and answer

Let λ be the left regular representation of a countable group Γ , let $\xi = \delta_e$, and let τ be the regular trace. For a scalar positive definite function ϕ and finite nonempty $F \subset \Gamma$, put

$$\phi[F] = [\phi(g^{-1}h)]_{g,h \in F}.$$

Austin proves the general lower bound

$$\Delta\phi_{\text{ac}} \leq \inf_{\emptyset \neq F \in \Gamma} (\det \phi[F])^{1/|F|}. \quad (1)$$

On PDF page 36 of arXiv:2412.13751v4, Problem 3.7 asks:

Suppose that equality holds in (1) whenever $\phi = \tau(a^*(\cdot)a)$ for some $a \in \mathbf{C}[\Gamma]$. Must Γ be amenable?

See Figure 1.

Theorem 1 (Full answer to Problem 3.7). *Yes. If equality in (1) holds for every $\phi = \tau(a^*(\cdot)a)$ with $a \in \mathbf{C}[\Gamma]$, then Γ is amenable. In fact, it is enough to assume equality for the positive elements*

$$a = 1 + t \sum_{s \in S} s, \quad S = S^{-1} \in \Gamma \setminus \{e\}, \quad 0 < t < |S|^{-1}.$$

Difficulties of extension beyond amenable groups

It is natural to ask about generalizations of Theorem A to non-amenable groups or other C^* -algebras. Various directions suggest themselves, but none of them seems entirely straightforward. We quickly discuss four of these here. For simplicity, we now restrict our attention to scalar-valued positive definite functions and positive functionals.

First, Corollary 3.3 immediately gives the following: in the notation of Theorem A, if Γ is any countable group and $\varphi : \Gamma \rightarrow \mathbb{C}$ is positive definite, then

$$\Delta\varphi_{ac} \leq \inf \{ (\det \varphi[F])^{1/|F|} : F \subset \Gamma \text{ finite and nonempty} \}. \quad (3.8)$$

Theorem A shows that this is an equality if Γ is amenable. I suspect that this implication can be reversed, even if we allow only certain positive definite functions:

Problem 3.7. *Suppose that equality holds in (3.8) whenever $\varphi = \tau(a^*(\cdot)a)$ for some $a \in C[\Gamma]$. Must Γ be amenable?*

Figure 1: Equation (3.8) and Problem 3.7 in Austin, arXiv:2412.13751v4, PDF p. 36.

2 Proof intuition

The matrix $\phi[F]$ is a coordinate compression of a positive right-convolution operator B . Since every diagonal matrix coefficient of $\log B$ is the von Neumann trace, the logarithm of the ratio between the two sides of (1) is exactly the Jensen gap

$$|F|^{-1}(\text{Tr} \log(P_F B P_F) - \text{Tr}(P_F \log B P_F)).$$

The resolvent formula for $\log$ and a Schur complement show that this gap controls the Hilbert–Schmidt norm of the off-diagonal block $(1 - P_F)B P_F$.

Now prescribe a finite symmetric S and choose $a = 1 + t \sum_{s \in S} s$. The associated positive right-convolution operator is $T = 1 + tH$, and $B = T^2$. Positivity of T makes the Sylvester map $Y \mapsto CY + YA$ uniformly invertible, so leakage of T^2 controls leakage of H . But the columns of H are sums of distinct coordinate vectors; hence its leakage is exactly

$$\|(1 - P_F)H P_F\|_{\text{HS}}^2 = \sum_{s \in S} |Fs \setminus F|.$$

Therefore sharpness of the determinant infimum produces arbitrarily good Følner sets for every finite S .

3 A quantitative compression lemma

Lemma 2 (Compression gap for the logarithm). *Let B be a bounded positive invertible operator on a Hilbert space, let P be a finite-rank orthogonal projection, and regard PBP as an operator on $P\mathcal{H}$. Then*

$$\text{Tr}_{P\mathcal{H}} \log(PBP) - \text{Tr}_{P\mathcal{H}}(P \log(B) P) \geq \frac{\|(1 - P)BP\|_{\text{HS}}^2}{2\|B\|^2}. \quad (2)$$

Proof. Put $Q = 1 - P$, and write B in block form relative to $\mathcal{H} = P\mathcal{H} \oplus Q\mathcal{H}$:

$$B = \begin{bmatrix} A & X^* \\ X & C \end{bmatrix}, \quad A = PBP, \quad X = QBP, \quad C = QBQ.$$

For $r \geq 0$, the Schur complement formula gives

$$P(B+r)^{-1}P = (D_r - Y_r)^{-1}, \quad D_r = A + rP, \quad Y_r = X^*(C + rQ)^{-1}X.$$

Both D_r and $D_r - Y_r$ are positive invertible. If $Z = D_r^{-1/2}Y_r D_r^{-1/2}$, then $0 \leq Z < 1$ and $(1-Z)^{-1} \geq 1+Z$. Conjugating this inequality yields

$$(D_r - Y_r)^{-1} - D_r^{-1} \geq D_r^{-1}Y_r D_r^{-1}.$$

Set $M = \|B\|$. Since $C + rQ \leq (M+r)Q$ and $D_r \leq (M+r)P$, we obtain

$$\begin{aligned} \text{Tr}(P(B+r)^{-1}P - (A+rP)^{-1}) &\geq \text{Tr}(D_r^{-1}Y_r D_r^{-1}) \\ &\geq (M+r)^{-2} \text{Tr}(Y_r) \\ &\geq (M+r)^{-3} \|X\|_{\text{HS}}^2. \end{aligned}$$

The resolvent representation

$$\log x = \int_0^\infty \left(\frac{1}{1+r} - \frac{1}{x+r} \right) dr$$

now gives

$$\begin{aligned} \text{Tr} \log A - \text{Tr}(P \log(B)P) &= \int_0^\infty \text{Tr}(P(B+r)^{-1}P - (A+rP)^{-1}) dr \\ &\geq \|X\|_{\text{HS}}^2 \int_0^\infty (M+r)^{-3} dr = \frac{\|X\|_{\text{HS}}^2}{2M^2}. \end{aligned}$$

This is (2). □

4 From the determinant gap to a group boundary

For $s \in \Gamma$, let $R_s \delta_g = \delta_{gs}$ be right translation on $\ell^2(\Gamma)$. It commutes with λ . Fix a finite symmetric $S \subset \Gamma \setminus \{e\}$, put

$$H = \sum_{s \in S} R_s, \quad T = 1 + tH, \quad 0 < t < |S|^{-1},$$

and set

$$\alpha = 1 - t|S|, \quad \beta = 1 + t|S|.$$

Since $H = H^*$ and $\|H\| \leq |S|$, we have

$$0 < \alpha \leq 1 \leq T \leq \beta. \tag{3}$$

The group-ring element $a = 1 + t \sum_{s \in S} s$ is therefore positive and invertible, and its right-convolution operator is T (up to the harmless inverse convention for right translation).

Lemma 3 (Boundary lower bound). *Let $\phi = \tau(a^*(\cdot)a)$. For every finite nonempty $F \subset \Gamma$,*

$$\log \frac{(\det \phi[F])^{1/|F|}}{\Delta \phi} \geq \frac{2t^2(1-t|S|)^2}{(1+t|S|)^4} \frac{1}{|F|} \sum_{s \in S} |Fs \setminus F|. \tag{4}$$

Proof. Let $P = P_F$ be the coordinate projection onto $\ell^2(F)$ and $Q = 1 - P$. The vectors $\lambda(g)a\xi$, $g \in F$, are exactly $T\delta_g$, up to the same inverse convention. Thus their Gram matrix is

$$\phi[F] = PT^2P|_{P\ell^2(\Gamma)} \quad (5)$$

up to simultaneous transposition, which does not change the determinant. The source's definition gives

$$\log \Delta\phi = 2 \log \Delta T = \tau(\log T^2). \quad (6)$$

Because T^2 commutes with λ , every diagonal coefficient of $\log T^2$ is $\tau(\log T^2)$. Hence

$$\text{Tr}(P \log(T^2) P) = |F| \log \Delta\phi.$$

Applying Lemma 2 to $B = T^2$ and using $\|T^2\|^2 \leq \beta^4$ yields

$$\log \frac{(\det \phi[F])^{1/|F|}}{\Delta\phi} \geq \frac{\|QT^2P\|_{\text{HS}}^2}{2\beta^4|F|}. \quad (7)$$

It remains to control the off-diagonal block. Write

$$A_0 = PTP, \quad C_0 = QTQ, \quad Y = QTP.$$

Then $A_0, C_0 \geq \alpha$ by (3), and block multiplication gives

$$QT^2P = C_0Y + YA_0.$$

With the Hilbert–Schmidt inner product,

$$\begin{aligned} 2\alpha\|Y\|_{\text{HS}}^2 &\leq \text{Tr}(Y^*C_0Y) + \text{Tr}(Y^*YA_0) \\ &= \langle Y, C_0Y + YA_0 \rangle_{\text{HS}} \leq \|Y\|_{\text{HS}} \|QT^2P\|_{\text{HS}}. \end{aligned}$$

Consequently

$$\|QT^2P\|_{\text{HS}} \geq 2\alpha\|QTP\|_{\text{HS}} = 2\alpha t\|QHP\|_{\text{HS}}. \quad (8)$$

For a fixed $g \in F$, the vectors δ_{gs} , $s \in S$, are distinct. After projecting outside F , they remain orthogonal. Summing the squared column norms gives the exact identity

$$\|QHP\|_{\text{HS}}^2 = \sum_{g \in F} \#\{s \in S : gs \notin F\} = \sum_{s \in S} |Fs \setminus F|. \quad (9)$$

Combining (7), (8), and (9) proves (4). $\square$

5 Completion of the proof

Assume the hypothesis of Theorem 1. Fix an arbitrary finite symmetric $S \subset \Gamma \setminus \{e\}$ and any $0 < t < |S|^{-1}$. For the corresponding a and ϕ , equality in (1) says that there are finite nonempty sets F_n for which

$$\log \frac{(\det \phi[F_n])^{1/|F_n|}}{\Delta\phi} \longrightarrow 0.$$

Lemma 3 and positivity of its constant imply

$$\frac{1}{|F_n|} \sum_{s \in S} |F_n s \setminus F_n| \longrightarrow 0.$$

In particular $|F_n s \setminus F_n|/|F_n| \rightarrow 0$ for every $s \in S$. Since $S = S^{-1}$, this is equivalent to $|F_n \triangle F_n s|/|F_n| \rightarrow 0$ for every $s \in S$.

Every finite subset of Γ is contained in a finite symmetric one after adjoining inverses. We have therefore established the Følner criterion for Γ , so Γ is amenable. This proves Theorem 1.

Equivalently, the contrapositive is quantitative. If Γ is nonamenable, then some finite symmetric S has

$$\inf_{\emptyset \neq F \in \Gamma} \frac{1}{|F|} \sum_{s \in S} |F s \setminus F| > 0.$$

For every $0 < t < |S|^{-1}$, Lemma 3 supplies a uniform strict gap in (1) for the single element $a = 1 + t \sum_{s \in S} s$. This also gives a concrete relation to the isoperimetric profile requested in Problem 3.8 of the source.

6 Verification, scope, and novelty

The proof uses only positive invertible group-ring elements, so no kernel, extended-logarithm, or singular-part convention is involved. Equations (5) and (6) match the source’s Radon–Nikodym operator and determinant normalization: for a scalar coefficient represented by $T\xi$, the source defines $\Delta\phi = (\Delta T)^2$.

The script `code/compression_gap_probe.py` checks Lemma 2 on 300 random complex positive matrices, the Sylvester estimate (8) on another 300 random positive matrices, and the boundary identity on a finite cyclic model without wraparound. These checks passed. They are regression evidence, not part of the proof.

The run’s four cheap indexes and local source corpus were searched for the arXiv id, title, and combinations of “amenability”, “finite compression”, and “Fuglede–Kadison determinant”. Bounded exact-phrase, title/citation, and arXiv web searches on July 21, 2026 found arXiv:2412.13751v4, last revised March 20, 2026, where the question remains Problem 3.7, but no later paper claiming a resolution. This is therefore a candidate full solution with moderate, not definitive, novelty confidence.

For human review, the decisive points are Lemma 2, the right-convolution identification in (5), and the normalization in (6). The argument does not claim a sharp constant in (4), nor does it classify all group-ring elements that witness nonamenability.

References

- [1] T. Austin, *Entropy and determinants for unitary representations*, arXiv:2412.13751v4, 2026.